

ABSTRACT

Energy consumption varies over time and is influenced by previous usage patterns, daily temporal behavior, indoor environmental conditions, and outdoor weather. This project presents a machine learning-based system for forecasting household energy consumption one hour ahead using the UCI Appliances Energy Prediction Dataset. The dataset contains 19,735 observations recorded at 10-minute intervals with 29 original variables covering appliance energy consumption, lighting, indoor temperature and humidity, and outdoor weather conditions. The data was processed chronologically by converting and sorting timestamps, validating data quality, removing redundant variables, and constructing a one-hour-ahead target using six 10-minute time steps. Feature engineering was performed to capture temporal and historical consumption behavior through time-based, cyclical, lag, rolling-statistical, weather, and weather-change features. The resulting data was divided chronologically into training, validation, and test sets to preserve the time-series order. Multiple approaches, including Persistence, 24-Hour Lag, Linear Regression, Random Forest, Gradient Boosting, XGBoost, and Ridge Regression, were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . Ridge Regression was selected after hyperparameter tuning using the validation set. The final model was retrained on the combined training and validation data and evaluated on an untouched test period, achieving an MAE of 39.63 Wh, an RMSE of 80.08 Wh, and an R^2 of 0.233. The forecasting pipeline was integrated into a Streamlit application supporting LIVE, LIVE SIMULATION, MANUAL PREDICTION, and HISTORICAL DATASET modes. The system demonstrates an end-to-end workflow from data preparation and feature engineering to model evaluation and interactive forecasting, while providing a foundation for future integration with smart meters, IoT devices, and automated real-time energy data sources.

Keywords: *Energy Consumption Forecasting, Time Series, Machine Learning, Feature Engineering, Ridge Regression, Weather Features, Streamlit*